NIRANJAN CHENNASAMUDRAM BALAJI

248-413-7304 niranjcb@gmail.com linkedin.com/in/niranjancb github.com/niranj1234

SUMMARY

Embedded Engineer with over 3 years of experience in automotive Electronic Control Units (ECU) validation, embedded software development, & data engineering. Skilled in Embedded C/C++, Python, Hardware-in-Loop (HIL) testing, automation, communication protocols, and cloud platform.

EDUCATIONAL QUALIFICATIONS

Master of science: Embedded System Engineering Apr 2024
Oakland University, Rochester Hills, Michigan, USA
Bachelor of Engineering: Electronics and Communication Engineering Nov 2020
Anna University, Chennai, Tamil Nadu, India

SKILLS

Programming Languages: Embedded C/C++, Python, JavaScript, SQL
Embedded Tools: CANalyzer, D-Space Control Desk, Automation Desk, STM32Cube IDE
Hardware: Artix A-7, KL25Z, Arduino UNO, STM32F4DISCOVERY | **Communication Protocols**: CAN, UART, SPI, I2C, TCP/IP
Cloud & Data: AWS S3, Athena, Databricks, Informatica Powercenter, Oracle
Testing Framework: Pytest, Gtest, Jest based E2E testing.
Operating Systems: Linux, Windows, Mac | **Version Control**: Git, Github | **Defect Management & Issue Tracking**: Jira
Certificate: Microcontroller Embedded C Programming: Absolute Beginners, Embedded Systems Bare-Metal Programming Ground Up (STM32), FreeRTOS From Ground Up on ARM Processors, Simulink Onramp, MATLAB Onramp

EXPERIENCE

Embedded Software Engineer Nov 2024 - Apr 2025
Cognizant Technology Solutions (Meta Reality Lab) - Redmond, WA

- Implemented unit tests in C/C++ using Google Test (GTest) and developed end-to-end (E2E) tests in JavaScript using Jest to validate build information, hardware-firmware interaction and overall system stability.
- Worked closely with firmware and hardware teams to debug boot issues and trace failures back to source-level problems, helping improve stability across different chip variants.
- Developed and executed automation test scenarios for multicore MCU boot validation, leveraging core dump analysis and log verification to ensure stable boot sequences in CI/CD pipelines.

EE Validation Engineer Sept 2023 - Oct 2024
Cognizant Technology Solutions (Stellantis) - Auburn Hills, MI

- Performed functional and diagnostics validation of ECUs (e.g., BCM, Instrument Cluster) using CANalyzer, CDA, dSPACE ControlDesk, and AutomationDesk in vehicle and HIL setups, ensuring adherence to vehicle specifications.
- Found and reported issues from failed test cases, then analyzed logs to help the development team figure out and fix the root cause.
- Debugged and reworked dSPACE model code written in C to update system behavior, fix simulation issues, and ensure testbed accuracy.
- Reduced manual test effort by Designing and implementing automated test scripts using Python to verify the functionality of multiple ECU's.
- Support all build phases, including topology design, gateway schematic development, and ECU translations, to ensure seamless communication between components.

Engineering Intern May 2023 – Aug 2023
Adduxi Inc - Rochester Hills, MI

- Responsible for ensuring the functionality and performance of the manufacturing robot.
- Assisted in repairing the robots and programming in PLC.

Data Engineer May 2021 to Aug 2022
Infosys (Toyota) - Chennai, India

- Developed and implemented data migration processes to transfer data from Oracle databases to Amazon Web Services.
- Analyzed Informatica mappings & workflows to understand data logic & storage requirements, & subsequently developed solutions for efficient data transfer to S3.
- Implemented CI/CD pipelines using Jenkins, where each deployment involved pushing a .config file and a JavaScript deployment script to GitHub. Jenkins pulled both files and executed the script to deploy assets to AWS S3 as per the config.
- Developed Python automation to trigger data migration scripts and config files on a scheduled basis, ensuring consistent and error-free daily uploads to AWS S3.

ACADEMIC PROJECT

Step Impediment Detection (SID) : Developed a visual impairment assistance device to detect obstacles while walking. Used an FPGA (Artix-A7 board) for detection, and a handheld device provided feedback.

Autonomous Car : Developed an autonomous car featuring ADAS capabilities, including adaptive cruise control, forward collision control, and speed reduction, using Arduino, PIXY2, IR sensor, and ultrasonic sensor.